Bryant Har

Q2a) Describe in a 1 page summary the major findings reported in the 1991 paper.

In the 1991 paper, the authors describe the design and advantages of a fully depleted lean channel transistor (DELTA) with its gate incorporated into a new vertical ultra-thin SO1 structure. The study highlights its advantages over conventional planar MOSFETs, especially in addressing challenges at deep submicron scales (0.3–0.1 μm). The DELTA structure is fabricated using selective oxidation, producing a high-quality single-crystal SOI channel without defects. In this deep-submicrometer region, selective oxidation produces and isolates an ultra-thin SO1 MOSFET that has high crystalline quality. This quality is typically as good as that of conventional bulk single-crystal devices. This design and fabrication process simplifies integration with existing MOSFET fabrication processes while retaining high compatibility. Experiments and three-dimensional simulations have shown that structure has effective channel control, and that the vertical ultra-thin SO1 structure provides superior device characteristics. Specifically, the authors point out reduction in short-channel effects, minimized subthreshold swing, and high transconductance. The DELTA design demonstrates strong immunity to SCE due to the buried SiO2 layer and enhanced gate control over the channel. The vertical ultra-thin SOI design achieves reduced subthreshold swing, improving on/off current ratios critical for low-power applications. The vertical design increases the effective channel width, enhancing transconductance and enabling higher drive current compared to traditional planar devices. The DELTA structure's gate envelops the channel, ensuring superior electrostatic control, which minimizes leakage and punchthrough effects while maintaining device performance even at low supply voltages. The paper concludes that the DELTA structure significantly enhances device performance and is a viable path forward for submicron transistor scaling.

Q2b) Write a 1/2 page summary of the Intel whitepaper on how the FinFET was taken from the laboratory to production and commercialization and why is it a true advance in the field.

Intel's transition of FinFET tri-gate technology from concept to production was a breakthrough in semiconductors. Originally proposed in 1991 as the "DELTA" structure, this 3D design offered solutions to the growing limitations of traditional planar transistors, such as poor control over electrical leakage and reduced performance at smaller scales. By 2011, Intel became the first company to bring FinFET-based 22nm chips to mass production, setting a new standard for efficiency and performance. The major advancement was the FinFET's 3D structure, where the transistor gate wraps around a silicon "fin". This improved the control of current flow, reduced power leakage, enabled faster speeds, and allowed smaller, denser chips without compromising performance. Compared to previous planar technology, Intel realized a 50% reduction in active power usage, making it a true advancement in the field and enabling far smaller chips to follow and setting a new benchmark in the performance of modern devices.

Q4)

Summarize the impact of heterostructures on semiconductor devices stated in the lecture.

Heterostructures have revolutionized semiconductor devices by introducing interfaces where the properties of two different materials meet. These interfaces enable unique functionalities by tailoring band alignments and controlling carrier dynamics in ways impossible with homostructures. Devices like heterojunction bipolar transistors (HBTs) leverage wide-bandgap emitters to minimize recombination and increase efficiency, while double-heterostructure lasers confine carriers and photons in a narrow active region, enabling continuous operation at room temperature. Such innovations have driven advancements in high-speed communications, efficient light sources, and cutting-edge optoelectronic systems. Even today, Kroemer's work on heterostructures has precipitated record-setting performance in new devices.

Write one more paragraph to update Kroemer's article to include new developments in semiconductor heterostructure devices since the writing of this article in 2000.

Since 2000, the heterostructures that Kroemer discussed have evolved significantly with advancements in strained-layer engineering and 2D materials. Strain engineering in silicon-germanium (SiGe) HBTs, for example, has enhanced carrier mobility and performance in RF applications. Additionally, van der Waals heterostructures using materials like graphene and transition metal dichalcogenides (TMDs) provide atomically sharp interfaces and tunable electronic properties. These advances, all spawned from Kroemer's work on heterostructures, have enabled devices such as high-electron-mobility transistors (HEMTs) with record-setting speeds and low power consumption, as well as quantum well and superlattice structures for precise energy confinement. These developments emphasize the continued impact of

heterostructures and HBT's founded on Kroemer's work on next-generation electronics and photonics.

Q5) Which technology will you choose and why? Write a concise 1-page report and include data and reasoning.

I would propose starting a quantum dots startup, since they're a new technology that has the potential to revolutionize multiple areas of application. Quantum dots are semiconductor nanocrystals that exhibit quantum confinement effects when their size approaches the material's exciton Bohr radius, typically 2-10 nm. This confinement leads to discrete, size-dependent energy levels and tunable bandgaps. This makes them well-suited for applications in single-electron transistors, solar cells, LEDs, lasers, quantum computing, cell biology research, and microscopy. Compared to previous technologies, quantum dots have high quantum yield (up to 90%), narrow emission spectra (FWHM ~25-35 nm), and broad absorption spectra. Common materials include II-VI semiconductors (CdSe, CdS, ZnSe) and III-V semiconductors (InP, InAs). The field itself is also rapidly evolving. New perovskite quantum dots have defect-tolerant electronic structures and quantum yields near 1. Another advancement is the molten-salt synthesis method, enabling growth of high-quality III-V nanocrystals previously unattainable in solution. Challenges persist in stability, with surface oxidation and aggregation being major issues. The toxicity of cadmium-based dots also remains a concern, but there is ongoing research into safer indium and copper alternatives. Similarly, scaling up manufacturing is also an ongoing challenge. Applications span displays (improving color gamut in LCDs), photovoltaics (multi-exciton generation, theoretical efficiency limit of 66%), bio-imaging (long-term image stability), and potentially quantum computing (spin qubits with long coherence times). This last area in particular is my area of interest and expertise. The field is rapidly evolving, with current research focusing on enhancing stability and improving synthesis methods for large-scale production.

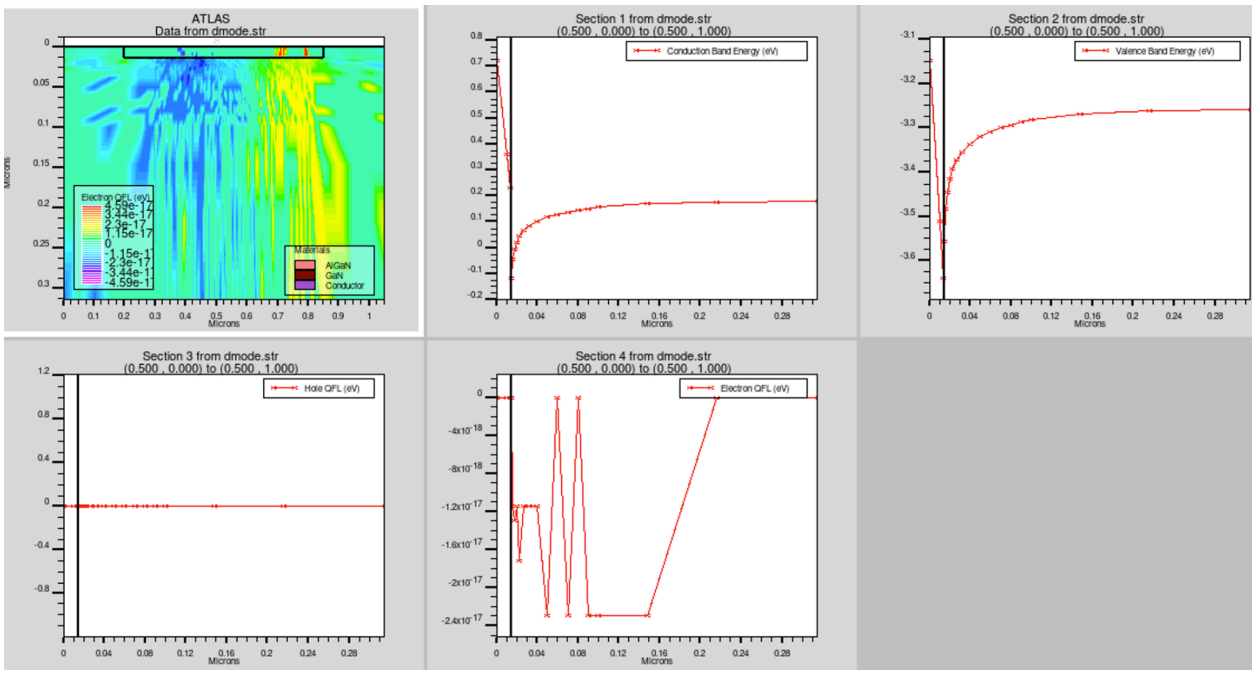
Q3)

d)

g)

D-Mode Band Diagram along the Cutline

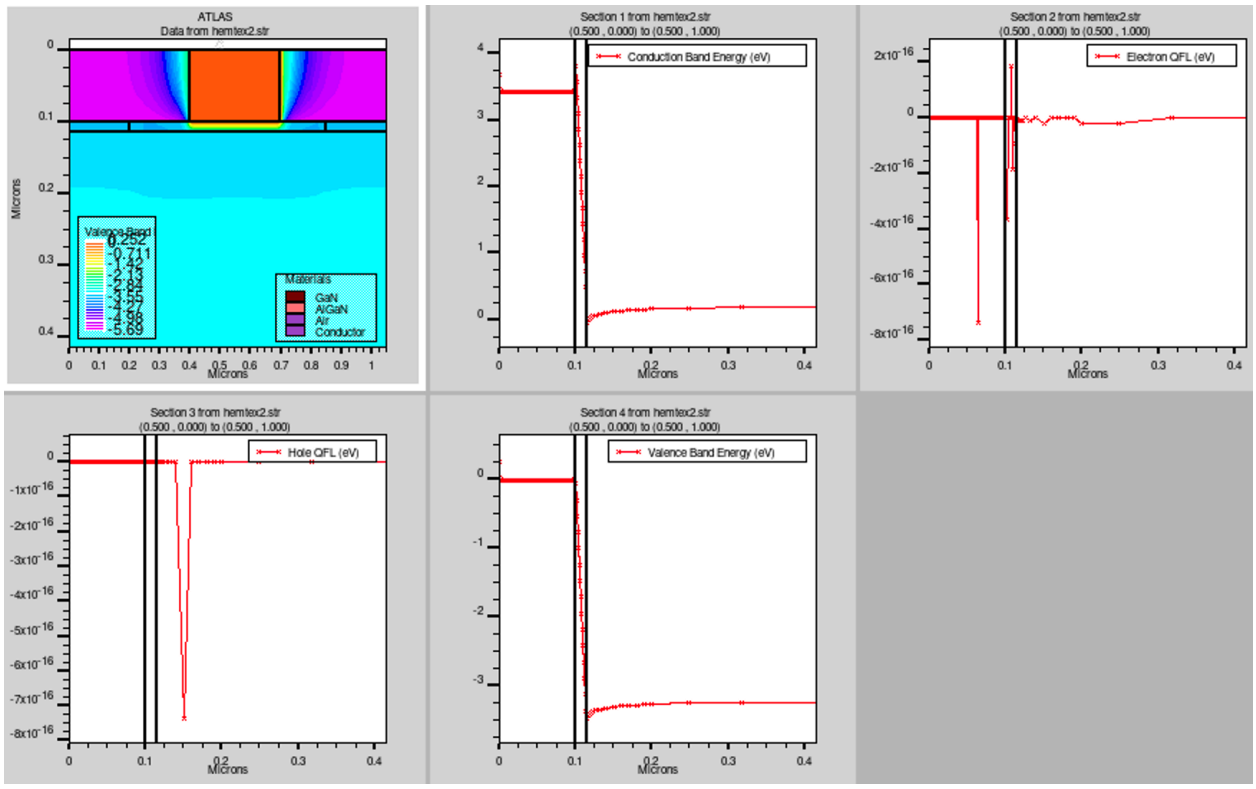
Conduction Band, Valence Band, Hole QFL, Electron QFL



h)

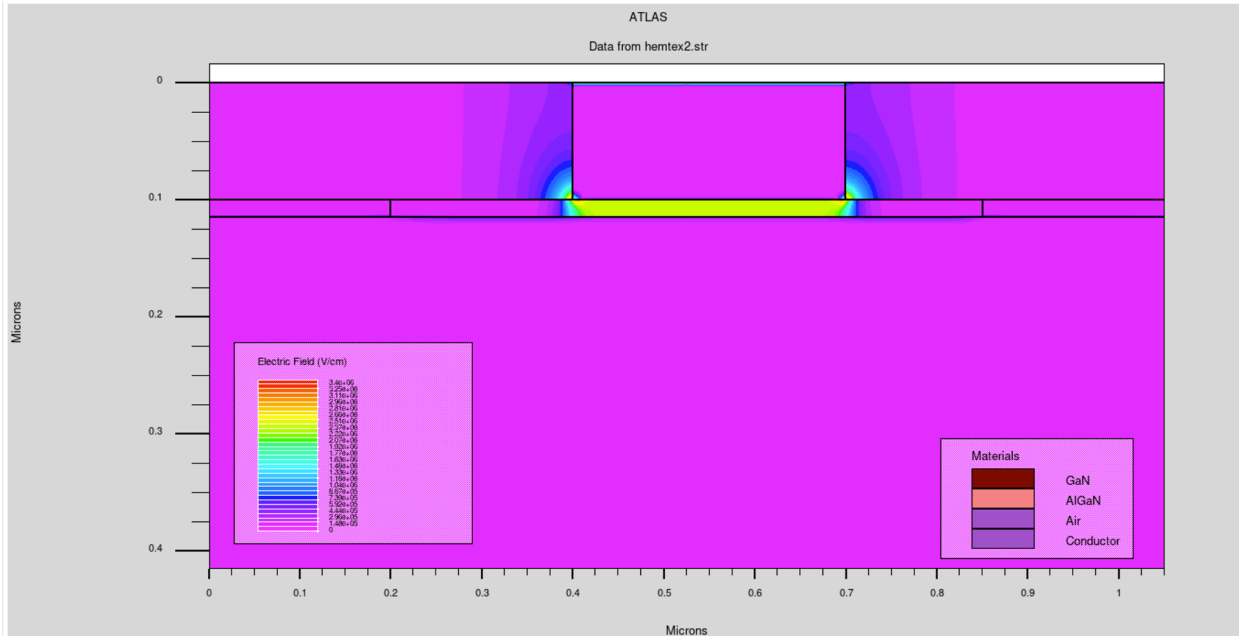
E-Mode Band Diagram along the Cutline

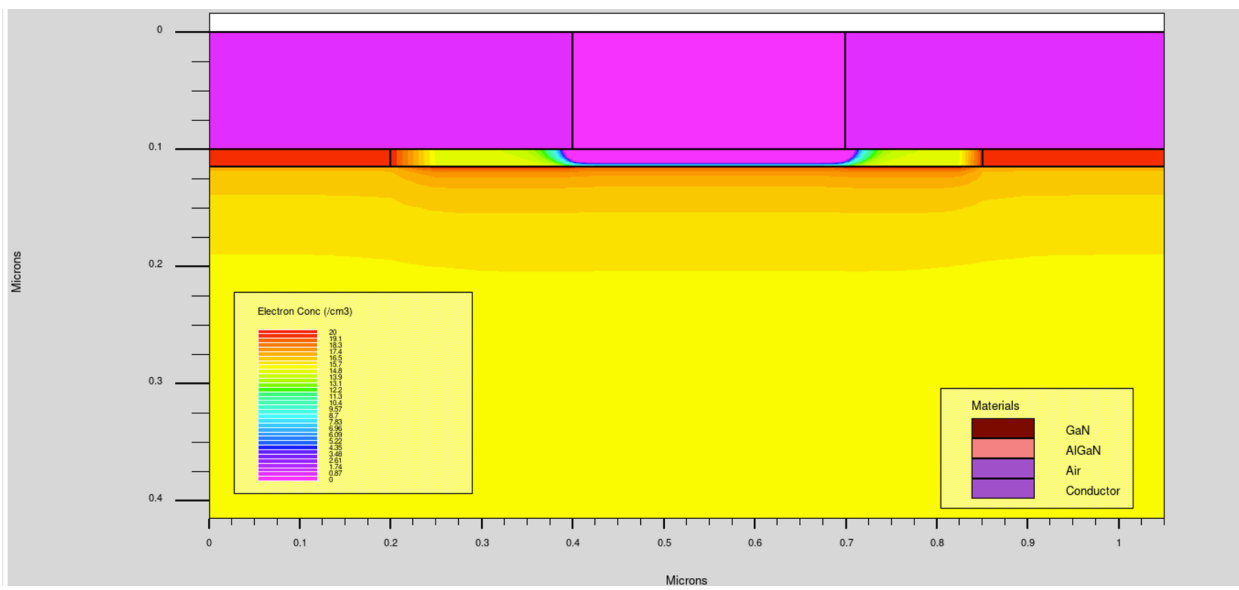
Conduction Band, Valence Band, Hole QFL, Electron QFL



i)

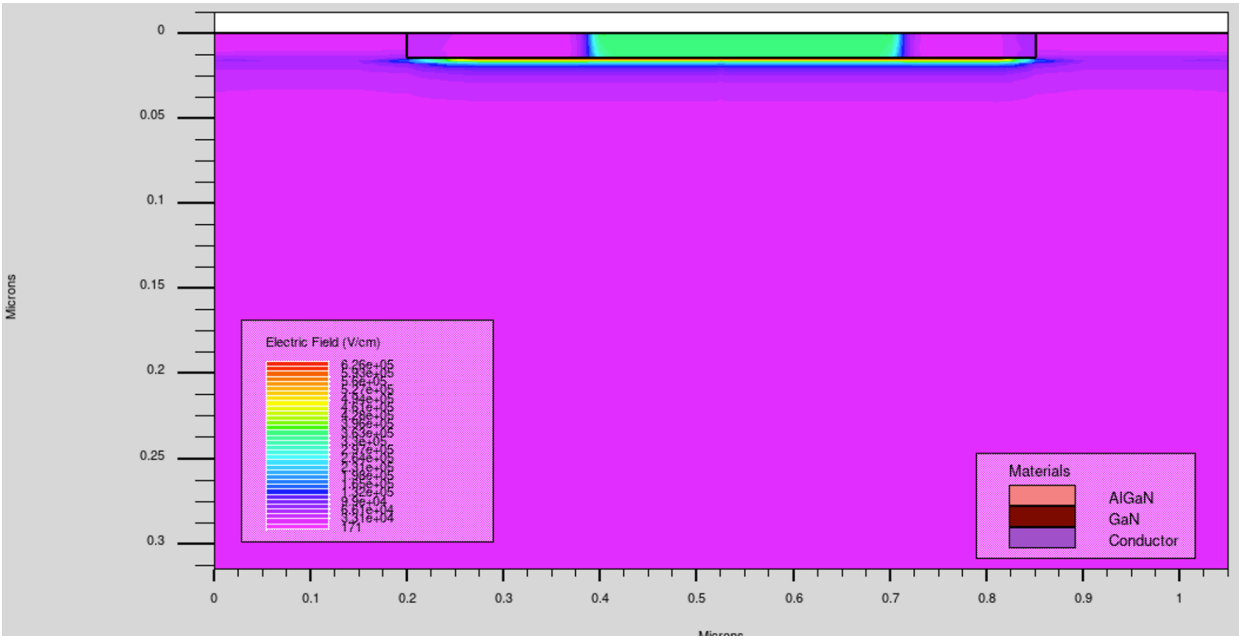
Electric Field / Electron Concentration (E-Mode)



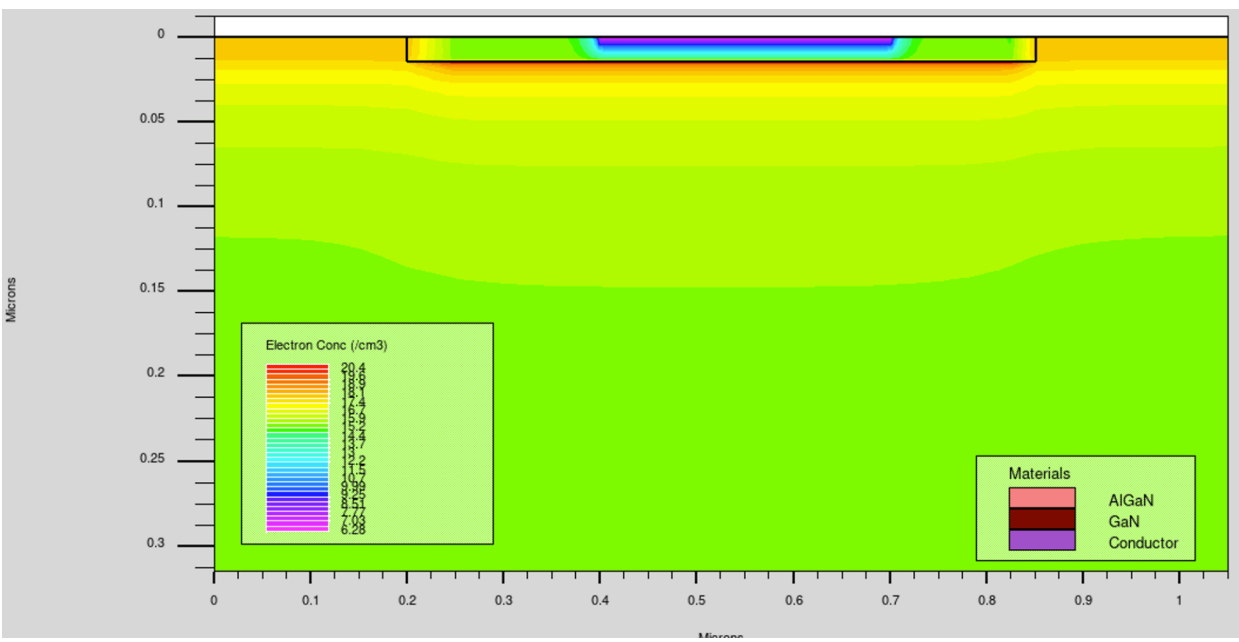


Yes the charts correspond to expectation. Explanation follows next charts.

Electric Field / Electron Concentration (D-Mode)



Electron Concentration (E-Mode)



Yes they match. We observe the results we saw earlier. There is a channel formed, and we observe a high electron concentration in the channel (Red in e conc chart).